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Urban resilience: A conceptual framework

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ABSTRACT

Nowadays, natural disasters and human actions are considered uncertain, unpredictable urban threats. Therefore, cities must respond more quickly and more effectively to anticipate and minimize the associated consequences and dangers. Promoting urban resilience in relation to environmental, socioeconomic and political domains has increasingly attracted the attention of researchers and local authorities. Thus, this article reviewed the scientific and technical literature about urban resilience highlighting its definitions, dimensions, application areas, characteristics and challenges and opportunities. Considering these topics, the aim of this work was to create a systematic approach and a clear view about urban resilience for building and strengthening cities against new disturbances. It was concluded that urban resilience is based on four basic pillars: resisting, recovering, adapting and transforming. Urban resilience can also be divided in five dimensions: natural, economic, social, physical and institutional. Additionally, urban resilience evaluation models integrate eleven characteristics: redundancy, robustness, connectivity, independence, efficiency, resources, diversity, adaptation, innovation, inclusion and integration. Finally, it was concluded that there are a lack of tools and methods to evaluate resilience providing a main challenge and opportunity for developing future holistic studies about urban resilience, allowing the identification of the most effective measures to increase resilience of different urban systems.

1. Introduction

One problem facing society today is the growth of large cities, with a significant increase expected in the coming decades. In 1900, only 10% of the world's population resided in urban areas (Grimm et al., 2008), with a higher level of urbanization since 1950 (Jedwab, Christiaensen, & Gindelsky, 2017). Today, the world's population is 7.6 billion and anticipated to reach 9.8 billion in 2050. In addition, the population living in urban areas is expected to increase from 55% to 68% by 2050 (United Nations, 2017, 2018).

The urbanization process is a key-phenomenon in the economic development of cities, regions and even countries. It can be observed that it is associated with a significant concentration of human resources, economic activities and consumption of resources in cities. However, while covering only about 2% of the planet's surface, cities account for about 75% of the world's resource consumption (Madlener & Sunak, 2011).

Cities will continue to face new and continuing challenges for job creation, increased food security, provision of essential services (e.g. housing, clean water, sanitation, basic health services and education), green space planning and maintenance, and urban waste and wastewater management (Dubbeling, Campbell, Hoekstra, & van Veenhuizen, 2009; Vörösmarty, Green, Salisbury, & Lammers, 2000) as a result of the social, economic and political forces embedded in this growth machine (Elkhalifa, 2016; Molotch, 1976). Associated with these challenges are often the high prices of basic necessities such as food and fuels that result from the scarcity of production considering demand (population growth), which can generate riots and social unrest (Custot et al., 2012). Therefore, policy responses should be linked to local conditions and provide tools and instruments to achieve sustainable and resilient urbanization and maximize the benefits of agglomeration economies and minimize negative impacts and externalities (Zhang, 2016).

Considering the continuous increase of cities in size, population diversity and complexity, it can be observed that their vulnerability to future extreme events and disasters increases proportionally (Borden, Schmidtlein, Emrich, Piegorsch, & Cutter, 2007), which are potentially derived from changes in climatic conditions, tsunamis, earthquakes, hurricanes, among others (Hunter, 2005).

The projected impacts of changing climatic conditions along with current and future urbanization dynamics put even more pressure on urban communities and their respective government institutions

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(Custot et al., 2012). Considering this, tackling natural disasters is a key challenge for cities and their communities (Tumini, Villagra-Islas, & Herrmann-Lunecke, 2017).

However, despite the large number of major disasters that have occurred in recent years, few have threatened the survival of urban areas. As such, cities must become smarter to deal with this series of challenges arising from urbanization (Naphade, Banavar, Harrison, Paraszczak, & Morris, 2011). Therefore, improvements in the conditions underpinning urban resilience will help recover the impacts inherent to disasters (Jabareen, 2015; Rose, 2011).

The term resilience was introduced in scientific research, initially associated with ecological systems by Holling (1973, p.17): *"Resilience determines the persistence of relationships within a system and is a measure of the ability of these systems to absorb changes of state variables, driving variables, and parameters, and still persist. In this definition, resilience is the property of the system and persistence or probability of extinction is the result."*

The main aim of resilience is to reduce the impacts resulting from a disturbance, a concept transversal to various research areas with very similar definitions. However, there is no single definition applicable in the various areas, and several actions are associated to the definition of resilience such as reconfiguration (Martin, 2012), speed (Pimm, 1984), recovery (Bocchini, Frangopol, Ummenhofer, & Zinke, 2014; Carver, 1998), maintenance (Allenby & Fink, 2005; Holling, 1973), adjustment (Gao, Barzel, & Barabási, 2016) and resistance (Haimes, Crowther, & Horowitz, 2008; Zhou, Wang, Wan, & Jia, 2010). Therefore, it can be concluded that resilience is defined as the ability of a system to resist and/or adapt to a particular disturbance and recover its normal functioning or state of balance, which may set the initial baseline or a new situation.

Improving city resilience is an important need for urban communities, especially for territories that are most at risk. Although natural or man-made disasters are always tragic, post-disaster recovery and reconstruction can provide a unique opportunity for societies to solve serious structural problems and prevent future suffering for affected populations, leading to increased resilience of local communities (Schneider, 2002; Tumini et al., 2017).

Thus, resilience can be achieved through a bottom-up process with city residents taking the lead in planning and governance issues, forming strong social networks that manifest and reinforce local identity (Ercoskun & Ozuduru, 2014; Gunduz, Oner, & Knox, 2016). This resilience is increasingly used in a positivist and indiscernibly normative way in urban communities (Christophe, Andrew, Mark, Martina, & Rachel, 2014), and is increasingly integrated into political and academic structures (Cutter, Burton, & Emrich, 2010; Marc, 2013). As a result, socially innovative actions and initiatives are considered a primary source of resilience through creativity among communities and stakeholders (government and private institutions) (Mehmood, 2016).

The urban communities and government institutions with a coherent land use plan and risk mitigation strategies are able to build urbanizations that will be resilient to natural disasters, able to recover rapidly from these events and be able to adapt for many years at a low cost (Burby, Deyle, Godschalk, & Olshansky, 2000). According to Wang and Xue (2019), the literature on urban resilience have rapidly increased in the last decade, having exploded in both academic research and policy statements with the unprecedented urbanization process. However, there is a need for further research related to spatial morphology and urban spatial structures (Masnavi, Gharai, & Hajibandeh, 2019).

If the institutions responsible for urban management intend to promote resilience to avoid possible losses, several concepts and characteristics of resilience must be clearly defined. Therefore, this article will present the framework of concept of urban resilience and its characteristics based on an extensive literature review about the subject. In addition to identify and describe the main characteristics of urban resilience, it will be presented the dimensions of urban resilience, its main areas of research and finally its main objectives and opportunities for urban resilience. Thus, the paper aims to create a conceptual framework based on existing literature to support the production of urban resilience tools, in order to prepare and develop strategies and plans to improve city planning and management against disruptive events, which can affect the functioning of the city.

The remainder of this article has the following structure, Section 2 briefly describes the methodology used in this research. In Section 3, the urban resilience is contextualized, and the various definitions are presented. In Section 4, the main dimensions of the resilience are described. In Section 5, the main characteristics of the resilience are described. In Section 6, the main challenges and opportunities of urban resilience are presented. Finally, in Section 7, the main conclusions of the investigation are drawn.

2. Research methodology

To carry out a comprehensive review of the literature on urban resilience, a systematic process was applied to select the revisited articles. The procedure comprises three phases: (i) search based on online databases, (ii) screening process and (iii) refining and final analysis (Tukamuhabwa, Stevenson, Busby, & Zorzini, 2015). For the systematic selection of articles for review in this work, three databases were used: Web of Science, Google Scholar and Web of Knowledge, which are the most comprehensive databases for academic research (Hosseini, Barker, & Ramirez-Marquez, 2016). Moreover, Google was used as there are many technical documents with special relevance to this topic from entities that apply this concept in cities. To carry out the search in these databases, terms such as "Urban resilience", "Resilient urban systems", "Resilient cities" and "Resilience in cities" (as well as other associated terms) were used as research keywords. While searching for the articles and documents in the databases, there was a high number of results. Thus, for the scientific articles, the period related to the last 15 years of research on this subject was selected, that is, between 2003 and 2018 where the articles were sorted by relevance and articles with more than 10 citations were considered due to their relevance. Regarding the technical documents, those from the entities with the greatest impact on the issues related to cities and urban environments were collected.

The screening process was carried out in two stages to ensure the quality and relevance of the articles found. Initially only articles published in international journals indexed to the Web of Knowledge and/or Scopus database were considered, since the revision process is more rigorous and the level of acceptance from the scientific community is more guaranteed (Bergström, van Winsen, & Henriqson, 2015), as well as technical documents of entities with special importance. In the second step, titles, keywords and abstracts were checked to ensure that the articles were relevant to the study. After completing these two steps, 101 scientific articles were obtained.

In the last step, the articles found an improved form through the content review. This is important because in some papers and documents, resilience was considered only as a subtopic or just as a label where more efforts were made on other topics such as system security risk management and disaster response. Reports on the evaluation and characterization of urban resilience have also been considered, where the most relevant documents are selected. After all these steps, the result produced 83 articles. Finally, these articles were analyzed in depth in terms of the definitions of urban resilience, their characteristics and quantification dimensions used.

3. Definition of urban resilience

There is a vast body of research about the concept of urban resilience with the objective of characterizing and evaluating the concept of urban resilience by identifying the type of resources and characteristics that a resilient urban system must have. Thus, there are several similar definitions of the concept of urban resilience, although the

Table 1
Definitions of urban resilience.

Scientific Area(s)	Definitions of urban resilience	Author(s)
Agricultural and biological sciences;	Resilience is the degree to which cities tolerate the change before reorganizing around a new set of structures and processes and depends on the ability of cities to maintain their eco-systemic and human functions simultaneously.	Alberti et al. (2003)
Engineering;	A resilient city is a sustainable network of physical systems and human communities.	Godschalk (2003)
Agricultural and biological sciences;	It is recommended that resilience only be used in a restricted sense to describe specific system attributes relating to: (i) the amount of disturbances a system can absorb and remain within the same state or domain of attraction and (ii) that the system is able to self-organize.	Klein et al. (2003)
Environmental science; social sciences;	Resilience is the ability of a system to adjust to changing conditions.	Pickett et al. (2004)
Social sciences;	Urban resilience is the ability of a city to recover from destruction.	Campanella (2006)
Business management and accounting; psychology;	Urban resilience refers both to design changes (structural, architectural, spatial planning) and to management and governance measures that aim to prevent or mitigate the physical and social vulnerability of urban areas, to protect life, property and the economic activity of the city.	Coaffee and O'Hare (2008)
Environmental science;	Resilience is the ability of a socio-ecological system to sustain a given set of ecosystem services in the face of uncertainty and change for a community.	Ernstson (2008b)
Environmental science; social sciences;	Urban resilience means extending the concept of resilience from technical systems to social systems, particularly to cities, and their ability to recover and continue to provide their main functions of life, commerce, industry, government, and social gathering in the face of calamities and other hazards.	Hamilton (2009)
Environmental science; social sciences;	From a resilient perspective, governance can be thought as a propositional collective action to sustain and improve a regime, or to trigger a transition from the system to a preferable regime.	Ernstson et al. (2010)
Environmental science;	A resilient system is a system that can tolerate disturbances by means of characteristics or measures that limit its impacts, reducing or neutralizing damages and disturbances, and allowing the system to respond, recover and adapt quickly to such disturbances.	Wardekker et al. (2010)
Environmental science; social sciences;	Resilience is the ability of systems to organize and recover from changes and disruptions without change to other states-that is, systems that are "safe to fail."	Ahern (2011)
Business management and accounting; energy; engineering; social science;	Urban resilience usually refers to the ability of a city or urban system to resist a wide range of shocks and tensions.	Leichenko (2011)
Earth and planetary sciences; social sciences;	In the case of urban adaptation to climate, a resilience-based approach encourages practitioners to consider innovation and change to help recover from tensions and shocks that may or may not be predictable.	Tyler and Moench (2012)
Engineering; social sciences;	Urban resilience to flooding is a city's ability to tolerate flooding and reorganize if physical damage and socio-economic disturbances occur to prevent death and injury and maintain current socio-economic identity.	Liao (2012)
Environmental science; social sciences;	Although urban resilience usually refers only to the ability to maintain functions and structures, it must be framed in the visions of resilience (system persistence), transition (incremental system change), and transformation (system reconfiguration).	Chelleri (2012)
Social sciences;	Resilience is the ability of an urban asset, location and / or system to provide predictable performance.	Brugmann (2012)
Business management and accounting; social sciences;	A climate-resilient city is one that can resist climate stress, respond effectively to climate-related risks, and quickly recover from residual negative impacts.	Henstra (2012)
Business management and accounting; social sciences;	Resilience is the ability of an individual, community, or institution to respond dynamically and effectively to changing climatic conditions, continuing to function at an acceptable level.	Brown et al. (2012)
Environmental science; social sciences	A city resilient to disasters can be understood as a city that can successfully support measures to strengthen individuals, communities and institutions to: (a) reduce or avoid and future risks; (b) reduce current and future susceptibility to resist risks; (c) establish mechanisms and functional structures for disaster response; and (d) establish functional mechanisms and structures for disaster recovery.	Wamsler, Brink, and Rivera (2013)
Social sciences;	Resilience is the ability to resist and recover from disruptive challenges.	Coaffee (2013)
Energy; engineering; social sciences;	Resilience in cities usually refers to the ability to absorb, adapt, and respond to changes in an urban system.	Desouza and Flanery (2013)
Agricultural and biological sciences;	Resilience is the ability of a city to absorb disturbances while maintaining its functions and structures.	Lu and Stead (2013)
Business management and accounting; social sciences;	Resilience is the ability of populations and urban systems to resist a wide range of hazards and stresses.	Romero-Lankao and Gnatz (2013)
Agricultural and biological sciences; environmental science; social sciences;	Urban resilience is the ability of a city to persist without qualitative changes in its structure and function, despite the disturbances.	Wu and Wu (2013)
Engineering; social sciences;	A more comprehensive definition of a resilient city emphasizes a community's overall ability and ability to withstand stress, survive, adapt and recover from a crisis or disaster, and move forward quickly.	Wagner and Breil (2013)
Environmental science;	Resilience refers to a set of urban ecosystems that provide benefits to livelihoods and urban well-being.	McPhearson et al. (2015)
Social sciences;	Urban resilience refers to the ability of an urban system and all its socio-ecological and sociotechnical networks to maintain or rapidly return to the desired functions in the face of a disturbance and adapt to change, and to rapidly transform systems that limit capacity adaptive current or future.	Meerow, Newell, and Stults (2016)
Agricultural and biological sciences;	Urban resilience can be defined in evolutionary terms as a proactive vision for planning, policy formulation, and strategic direction in which communities play a vital role in resilient place modelling through their active learning ability, robustness, capacity for innovation and adaptability.	Mehmood (2016)
Engineering;	Resilience is the ability of individuals, communities, institutions, companies, and systems in a city to survive, adapt, and grow regardless of the type of chronic stress and acute shocks to which they are subject.	Spaans and Waterhout (2017)

concept of resilience is not completely clear and objective considering divergent urban contexts (Pendall, Foster, & Cowell, 2010).

As argued by Klein, Nicholls, and Thomalla (2003), the problem of resilience is the existence of a multiplicity of definitions and the transformation of any of them into operational tools. Table 1 presents the summary of the various definitions found in the analyzed literature to define urban resilience in different research areas.

The definitions presented for urban resilience, which are created from different perspectives, show that about half of the definitions are presented in a context of threat to the system, namely in terms of changes/alterations, e.g. of the climatic conditions, floods, among others. On the other hand, some definitions are presented in a context of urban sustainability, as well as a logic of systems change and transformation in order to improve service delivery capacities.

However, the main differences found in the definitions of resilience lie in actions and perturbations, i.e. if the system is able to tolerate, to maintain, to neutralize, to support, to persist, to resist, to absorb, to survive, to reorganize, to adjust, to adapt, to recover, to respond, to transform face a disturbance in the system (natural disasters, climatic conditions, calamities, crises or disruptive events) that create impacts (changes, stresses, destruction or uncertainties), may cause damage to the system.

After analyzing the presented definitions, four basic pillars of urban resilience were identified, **resisting**, **recovering**, **adapting** and **transforming**. Thus, Fig. 1 presents the flowchart that represents the concept of urban resilience, which translates the way the concept emerges and evolves into a conceptual tool where a resilient urban system will always have to be evaluated for its ability to resist, maintain, recover, adapt or transform in the face of a series of disturbances, whose impacts tend to create instability in the system's equilibrium.

Thus, taking into account all the definitions presented and the information in Fig. 1, it is concluded that urban resilience is the capacity of a city and its urban systems (social, economic, natural, human, technical, physical) to absorb the first damage, to reduce the impacts (changes, tensions, destruction or uncertainty) from a disturbance (shock, natural disaster, changing weather, disasters, crises or disruptive events), to adapt to change and to systems that limit current or future adaptive capacity.

3.1. Main application areas of urban resilience

According to the results presented in Table 1, it can be concluded that urban resilience has been applied in various different scientific areas that are related to the study and functioning of the different urban systems, demonstrating the scope of application of this concept. Thus, based on the literature analyzed, it is concluded that urban resilience is applied mainly in five research areas: **climatic condition changes**, **urban planning**, **urban communities**, **energy** and **disasters** (natural or human).

The main objective of all areas of research is to improve the quality of life of populations because they are the main drivers of the evolution of cities. It is important to highlight the role and intervention of communities in the operationalization of the concept of resilience. The area of research that has attracted the most attention from the scientific community is climatic condition changes because it is one of the main problems that cities will face soon (Wardekker, de Jong, Knoop, & van der Sluijs, 2010; Bahadur & Thornton, 2015; Gleeson, 2008; Goldner & Schmitz, 2010; Muller, 2007; Rafael et al., 2017; Tyler et al., 2016) together with externalities associated with population growth and the urbanization phenomena (Collier et al., 2013; Grimm et al., 2008; Suárez, Gómez-Baggethun, Benayas, & Tilbury, 2016; Wagner & Breil, 2013). Nonetheless, the study of urban resilience incorporating various approaches to resilience will make it possible to plan and transform cities and their urban systems in a more holistic and integrated way (Desouza & Flanery, 2013; Pickett, Cadenasso, & Grove, 2004).

4. Dimension of urban resilience

In the study of urban resilience, it is important to understand the dimensions of the process of characterization and evaluation of resilience. According to the literature, urban resilience analysis is significantly related to threats and risks associated with natural disasters. Patel and Nosal (2016) concluded that the dimensions that best characterize urban resilience are the dimensions adopted by Ostadtaghizadeh, Ardalan, Paton, Jabbari, and Khankeh (2015) for community resilience.

Ostadtaghizadeh et al. (2015) carried out a systematic review of evaluation models for disaster resilience in communities, summarizing ten models that try to measure this resilience and suggest that this concept be studied according to five dimensions: 1) **Physical**, 2) **Natural**, 3) **Economic**, 4) **Institutional** and 5) **Social**.

Thus, "physical" resilience includes the assignment of the resilience in infrastructures, "natural" resilience includes the ecological and environmental resilience, "economic" resilience includes the development of societies and economies, "institutional" resilience includes the governance and mitigation policies and "social" resilience includes

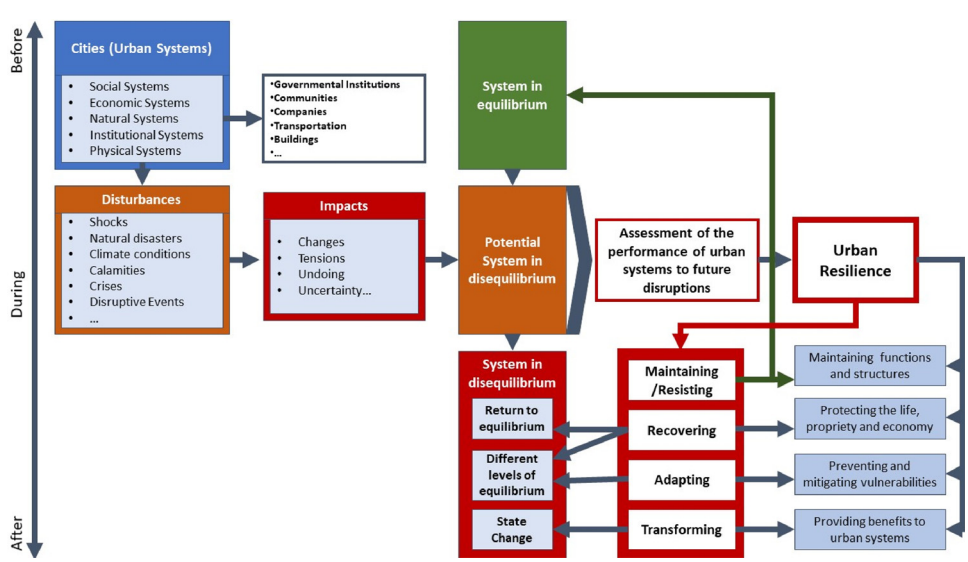

Fig. 1. Urban Resilience.

Table 2
Dimensions of urban resilience (Ostadtaghizadeh et al., 2015; Patel & Nosal, 2016).

Dimensions	Subdimensions	Author(s)
Physical	Physical Infrastructure, Infrastructural, Land Use and Structural Design	Campanella (2006), Muller (2007), Wamsler (2008), Shaw and Team (2009), Hamilton (2009), Wardekker et al. (2010), Ahern (2011), Liao (2012), Guo (2012), Collier et al. (2013), Desouza and Flanery (2013), Jabareen (2013), Sharifi and Yamagata (2014), Galderisi (2014), Tyler et al. (2016), Rafael et al. (2017), Spaans and Waterhout (2017), Dhar and Khirfan (2017) and Klein et al. (2017)
Natural	Ecosystems	Alberti et al. (2003), Nelson, Adger, and Brown (2007), Wallace and Wallace (2008), Ernstson (2008a), Shaw and Team (2009), Wardekker et al. (2010), Ernstson et al. (2010), Pelling and Manuel-Navarrete (2011), Pierce et al. (2011), Monteiro, Carvalho, Velho, and Sousa (2012), Brown et al. (2012), Tyler and Moench (2012), Allan, Bryant, Wirsching, Garcia, and Teresa Rodriguez (2013), Jabareen (2013), Collier et al. (2013), Gyasi et al. (2014), Sharifi and Yamagata (2014), Friend et al. (2014), Galderisi (2014), Lorenz and Lal (2015), Olazabal and Pascual (2016), Suárez et al. (2016), Tyler et al. (2016), Klein et al. (2017), Rafael et al. (2017), Spaans and Waterhout (2017), Carvalho, Martins, Marta-Almeida, Rocha, and Borrego (2017) and Dhar and Khirfan (2017)
Economic	Economic Development, Society and Economy	Pickett et al. (2004), Stanvliet and Parnell (2006), Muller (2007), Shaw and Team (2009), Pierce et al. (2011), Brown et al. (2012), Brugmann (2012), Collier et al. (2013), Cruz et al. (2013), Desouza and Flanery (2013), Jabareen (2013), Sharifi and Yamagata (2014), Galderisi (2014), Peyroux (2015), Toubin et al. (2015), Suárez et al. (2016), Tyler et al. (2016), Mehmood (2016), Spaans and Waterhout (2017) and Klein et al. (2017)
Institutional	Governance, Organized Governmental Services, Coastal Resource Management, Warning and Evacuation, Emergency Response, Disaster Recovery	Godschalk (2003), Pickett et al. (2004), Ernstson (2008a), Fleischhauer (2008), Gleeson (2008), Wallace and Wallace (2008), Coaffee and O'Hare (2008), Bogunovich (2009), Lamond and Proverbs (2009), Shaw and Team (2009), Gaillard (2010), Mortimer (2010), Hordijk and Baud (2011), Lang (2011), Leichenko (2011), Pelling and Manuel-Navarrete (2011), Pierce et al. (2011), Agudelo-Vera et al. (2012), Guo (2012), Henstra (2012), Brown et al. (2012), Campos and Zapata (2012), Tyler and Moench (2012), Vallance (2012), Friend and Moench (2013), Jabareen (2013), Lhomme et al. (2013), Lu and Stead (2013), Romero-Lankao and Gnatz (2013), Schmitt et al. (2013), Wagenaar and Wilkinson (2013), Wamsler et al. (2013), Zaidi and Pelling (2013), Coaffee (2013), Collier et al. (2013), Desouza and Flanery (2013), Galderisi (2014), Sharifi and Yamagata (2014), Vale (2014), Weichselgartner and Kelman (2014b), Bahadur and Thornton (2015), Jarvie et al. (2015), Lorenz and Lal (2015), McPhearson et al. (2015), Peyroux (2015), Toubin et al. (2015), Chelleri et al. (2015), Tyler et al. (2016), Dhar and Khirfan (2017), Klein et al. (2017), Spaans and Waterhout (2017), Vivonid et al. (2017), Ziervogel et al. (2017) and Leitner et al. (2018)
Social	Human Capital, Lifestyle and Community Competence, Society and Economy, Community Capital, Social and Cultural Capital, Population and Demographics Environmental, Risk Knowledge	Alberti et al. (2003), Pickett et al. (2004), Campanella (2006), Muller (2007), Nelson et al. (2007), Coaffee and O'Hare (2008), Ernstson (2008b), Wamsler (2008), Hamilton (2009), Bogunovich (2009), Gaillard (2010), Wardekker et al. (2010), Ahern (2011), Allan and Bryant (2011), Pierce et al. (2011), Agudelo-Vera et al. (2012), Guo (2012), McLellan et al. (2012), Monteiro et al. (2012), Tyler and Moench (2012), Vallance (2012), Brown et al. (2012), Collier et al. (2013), Desouza and Flanery (2013), Friend and Moench (2013), Jabareen (2013), Lhomme et al. (2013), Lu and Stead (2013), Schmitt et al. (2013), Allan et al. (2013), Wagenaar and Wilkinson (2013), Friend et al. (2014), Galderisi (2014), Gyasi et al. (2014), Pickett, McGrath, Cadenasso, and Felson (2014), Sharifi and Yamagata (2014), Béné, Cannon, Gupte, Mehta, and Tanner (2014), Chelleri et al. (2015), Jarvie et al. (2015), McPhearson et al. (2015), Peyroux (2015), Toubin et al. (2015), Mehmood (2016), Tyler et al. (2016), Dhar and Khirfan (2017), Klein et al. (2017), Rafael et al. (2017), Spaans and Waterhout (2017), Ziervogel et al. (2017) and Boix, Rausell, and Abeledo (2017)

resilience of communities and people in general. Table 2 presents the inherent set of reference works associated with the sub-dimensions identified in the literature for the five dimensions defined by Ostadtaghizadeh et al. (2015).

According to Table 2, for the five dimensions, several works and investigations can be framed in more than one dimension, and in some cases they are enclosed in the five dimensions (Collier et al., 2013; Galderisi, 2014; Jabareen, 2013; Sharifi & Yamagata, 2014; Tyler et al., 2016; Klein, Koenig, & Schmitt, 2017; Spaans & Waterhout, 2017). However, after the analysis in the literature the most discussed dimensions are institutional and social, also showing a strong link between them (Agudelo-Vera, Leduc, Mels, & Rijnaarts, 2012; Bogunovich, 2009; Brown, Dayal, & Rio, 2012; Chelleri, Waters, Olazabal, & Minucci, 2015; Coaffee & O'Hare, 2008; Collier et al., 2013; Desouza & Flanery, 2013; Dhar & Khirfan, 2017; Friend & Moench, 2013; Gaillard, 2010; Galderisi, 2014; Guo, 2012; Jabareen, 2013; Jarvie, Sutarto, Syam, & Jeffery, 2015; Klein et al., 2017; Leitner, Sheppard, Webber, & Colven, 2018; Lhomme, Serre, Diab, & Laganier, 2013; Lu & Stead, 2013; McPhearson, Andersson, Elmqvist, & Frantzeskaki, 2015; Peyroux, 2015; Pickett et al., 2004; Pierce, Budd, & Lovrich, 2011; Schmitt, Harbo, Diş, & Henriksson, 2013; Sharifi & Yamagata, 2014; Spaans & Waterhout, 2017; Toubin, Laganier, Diab, & Serre, 2015; Tyler & Moench, 2012; Vallance, 2012; Wagner & Breil, 2013; Ziervogel et al., 2017). This shows that a strategic and leadership structure is important because the municipal authorities are the main stakeholders. However, also effective leadership to guide government capacities for resilience in cities is an essential first step (Brown et al., 2012; Coaffee & O'Hare, 2008; Cruz, Costa, de Sousa, & Pinho, 2013; Ernstson et al., 2010; Godschalk, 2003; Klein et al., 2017; Leitner et al., 2018; Mehmood, 2016; Peyroux, 2015; Schmitt et al., 2013; Vale, 2014;

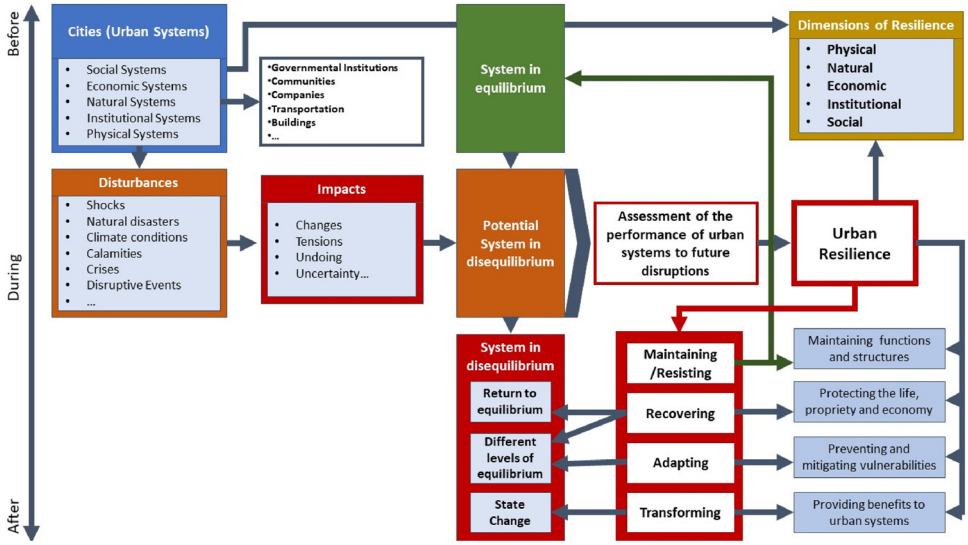

Fig. 2. Framing the dimensions of urban resilience.

Vivonid, Escalanteb, Baeza-Castroa, Mazari-Hiriartb, & Lernerb, 2017; Wagenaar & Wilkinson, 2013; Wallace & Wallace, 2008; Zaidi & Pelling, 2013).

Thus, it is possible from the construction of the concept to evolve the flowchart that involves the definition of the process of evaluation of Urban Resilience adding the various dimensions of resilience (Fig. 2) that derive from the conclusions drawn from the definitions analyzed in this Section 4.

5. Main characteristics of urban resilience

After presenting the concept and definitions of resilience, the main characteristics need to be identified that allow the evaluation of the resilience of urban systems, which in their genesis have a complex character where the technical components interact with the social ones.

After analyzing the various works, the following eleven most relevant characteristics to make an urban system more resilient were selected: **redundancy**, **diversity**, **efficiency**, **robustness**, **connectivity**, **adaptation**, **resources**, **independence**, **innovation**, **inclusion** and **integration** (Allan & Bryant, 2011; Godschalk, 2003; Kim & Lim, 2016; McLellan, Zhang, Farzaneh, Utama, & Ishihara, 2012; Spaans & Waterhout, 2017; Wardekker et al., 2010). Table 3 presents and describes a set of the main characteristics identified in the literature.

During the selection process and analysis of the different approaches to the concept of urban resilience, it was observed that several characteristics of different works had different denominations for the same concept, such as **buffering** and **robustness**, **omnivory** and **adaptation** (Wardekker et al., 2010), **social capital** and **resources**, **modularity**, **overlap in governance** and **redundancy**, **tight feedback** and **innovation**, **variability** and **adaptation** (Allan & Bryant, 2011), **interdependence**, **autonomy** and **connectivity** (Godschalk, 2003).

The definition of an urban system with characteristics of resilience presented in Table 3 will certainly make the application of the concept sufficiently operational for local stakeholders translating to concrete actions.

Adopting strategies to return urban resilience will allow us to develop useful and more robust approaches for cities to cope with and adapt to various types of disturbances, such as disruptions/blockages to the movement of populations. In this sense, local stakeholders should frame urban resilience as a highly flexible and adaptable approach, both in relation to a globally changing environment, and in relation to the specific situations and needs of the localities. This flexibility means that the system requires adaptation, in terms of creating appropriate formal (legal and governmental) frameworks and a different, more proactive approach to the local population (Godschalk, 2003; McLellan et al., 2012).

Fig. 3 is a result of an upgrade of Figs. 1 and 2 (where the concepts and definitions and dimensions of urban resilience were identified), illustrating the eleven characteristics identified in the literature associated with the evaluation of urban resilience, which must be considered in the implementation of measures and actions to promote a more resilient urban system.

6. Challenges and opportunities of urban resilience

Throughout the literature review on urban resilience, it could be observed that there are currently not many works/investigations that translate the development of tools to quantitatively evaluate the urban resilience. Most of the existing tools are summarized, particularly, to conceptual structures. This is reflected in the fact that most of the published studies about problems associated with urban issues/problems focus on the concept of sustainability and its evaluation at different scales rather than on the definition of resilience applied to cities and their urban systems (Sharifi & Murayama, 2013, 2014).

Weichselgartner and Kelman (2014a) presented two major concerns regarding the current resilience theory and application associated with the development of cities. First, many activities of building resilience in and for cities are based on assumptions about the social world, effectively imposing a basic framework, supported by more complex interrelated fields of action, values and meanings. Second, the current approach to a quantitative resilience city index is overgeneralized and fails to fully reveal the particularities of each city, hiding much more than it reveals, thus proving to be a limited indicator.

Although the concept of urban resilience points to a broad set of basic characteristics, there are few tools that incorporate them in an integrated way. Thus, this shortage of procedures and operational tools to evaluate the potential resilience of an urban system represents a gap in this research area, creating a challenge and opportunity for future work in this area.

Considering the analysis of the work about the urban resilience developed by several researchers, namely Sharifi and Yamagata (2014), Weichselgartner and Kelman (2014a, 2014b), and Spaans and Waterhout (2017), some key opportunities and challenges were identified in the area of urban resilience, such as:

Table 3
Main characteristics of urban resilience.

	Characteristics	Description	Author(s)
1	Redundancy	Existence of several functionally similar components, so that the system does not fail when one of the components fails.	Godschalk (2003), Allan and Bryant (2011), Wardekker et al. (2010), McLellan et al. (2012), Kim and Lim (2016) and Spaans and Waterhout (2017)
2	Diversity	Existence of several functionally different components to protect the system against the various threats. The more diversity the system possesses, the better the ability to adapt to a wide range of diverse circumstances.	Godschalk (2003) and Allan and Bryant (2011)
3	Efficiency	Positive relationship between the functioning of a static urban system in relation to the operation of a dynamic system.	Godschalk (2003) and Kim and Lim (2016)
4	Robustness	Ability to resists to attacks or other external forces. Robust design anticipates potential system failures, ensuring that failures are predictable, secure and not disproportionate to the cause.	Godschalk (2003), Wardekker et al. (2010), McLellan et al. (2012), Kim and Lim (2016) and Spaans and Waterhout (2017)
5	Connectivity	Connected system components for support and mutual interaction.	Godschalk (2003)
6	Adaptation	Ability to learn from experience and be flexible in the face of change.	Godschalk (2003), Wardekker et al. (2010), Allan and Bryant (2011), Kim and Lim (2016) and Spaans and Waterhout (2017)
7	Resources	Existence of resources that can be rapidly displaced to respond to disruptions and their effects.	Wardekker et al. (2010), Allan and Bryant (2011), McLellan et al. (2012), Kim and Lim (2016) and Spaans and Waterhout (2017)
8	Independence	Ability to operate for a continuous post-disaster period without relying on external physical intervention.	McLellan et al. (2012)
9	Innovation	Ability to quickly find different ways to achieve goals or meet their needs during a sock, or when a system is under stress. Innovation is critical to developing a city's ability to restore the functionality of critical systems under severely limited conditions.	Wardekker et al. (2010), Allan and Bryant (2011) and Spaans and Waterhout (2017)
10	Inclusion	Development of broad consultation and involvement of communities, particularly of the most vulnerable groups in the development of processes and plans. An inclusive approach contributed to a joint vision to build the city's resilience.	Godschalk (2003) and Spaans and Waterhout (2017)
11	Integration	Integration and alignment between urban systems promotes stronger decision-making and ensures that all users / components mutually support each other for a common outcome. The exchange of information between systems allows them to function collectively and respond quickly through shorter response cycles across the city.	Godschalk (2003) and Spaans and Waterhout (2017)

- Identification of criteria for the evaluation of urban resilience based on five dimensions: physical, natural, economic, institutional and social;
- Specialization in the management of urban resilience in the pre- and post-disaster phases based on the various characteristics of urban resilience;
- Use of a hierarchical scale of local needs classification that reflects the different compositions and needs in different contexts;
- Development of evaluation metrics that are easily scalable and replicable;
- Construction of conceptual frameworks that, although simple, are sufficiently robust to be used by non-specialists and suitable in self-evaluation procedures. These can consider the complexities and dynamics of cities as socio-ecological systems;
- Linking vulnerability, mitigation and adaptive capacity assessment measures to the conceptual frameworks for assessing urban resilience;
- Develop tools for assessing urban resilience of an iterative nature that fit each city;
- Continued need to re-examine planning approaches to urban resilience in relation to disruptions and interruptions/blockages in systems that have been or are expected to occur;
- Interpretation of urban resilience and the development of meaningful policy responses within a given space-cultural scenario, both from a technical perspective and the level of governance;
- Maintain urban resilience on the political agenda for a longer period, as well as extend interest and attention to the issue, particularly to cities and regions that (fortunately) do not have much experience with major disturbances/interruptions and impacts.

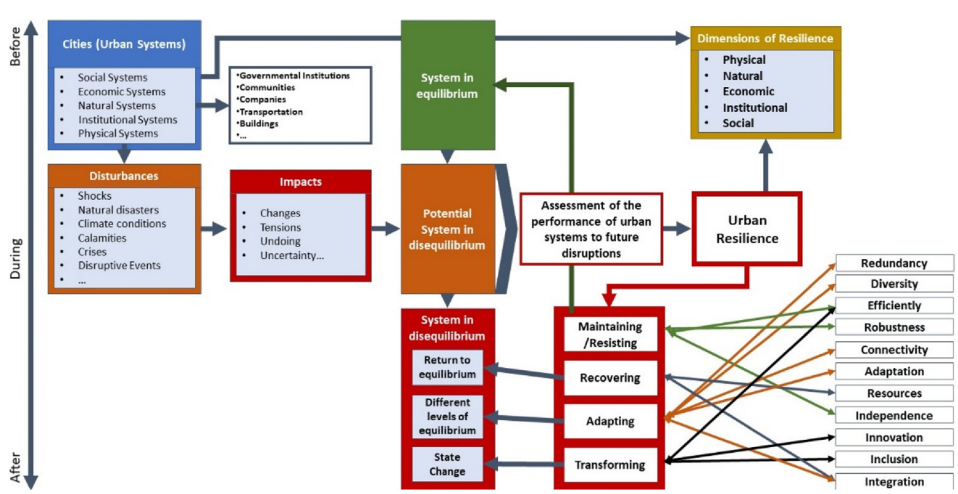

Fig. 3. Urban Resilience: framing of base characteristics.

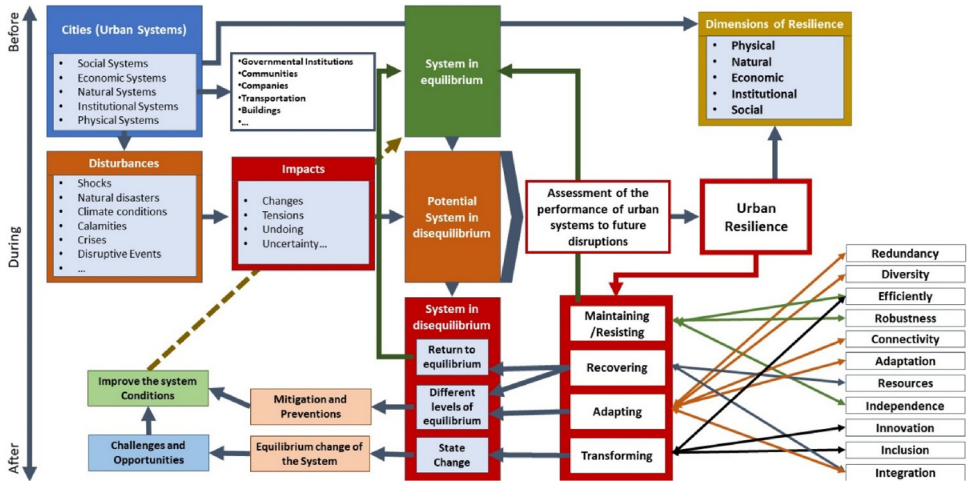

Fig. 4. Framing the Challenges and opportunities of the Urban Resilience.

The main starting point for addressing these challenges and embracing these opportunities is to combine various experiences, whether in or between cities. This includes the integration of different types of knowledge and a variety of approaches in which scientists, the public and authorities collaborate to generate not only scientifically reliable, but also contextual, socially robust and applicable knowledge. Thus, integrating diverse areas of knowledge, producing an explicit operational definition and concrete baselines, will overcome the current levels of "imprecision" that are associated with urban resilience at the conceptual level, as well as the disconnection locally (Weichselgartner & Kelman, 2014a).

Thus, a greater understanding is needed about which types of measures are effective in increasing urban resilience, and also how measures can be better designed, implemented and maintained to make cities, urban systems and their communities more resilient in terms of urban issues.

From the framework of the challenges and opportunities with the definitions (Fig. 1), dimensions (Fig. 2) and characteristics (Fig. 3) of urban resilience presented in the state of art, a diagram (Fig. 4) can be made that will underlie the development of an operational tool to evaluate urban resilience. Thus, considering a disturbance, an equilibrium urban system is affected and presents a wide range of impacts that can be measured and evaluated, thus creating a potential unbalance in the functioning of the systems. Moreover, it is important to understand and evaluate the level of urban resilience of these events in various dimensions. Thus, to evaluate the urban resilience is to verify if an urban system can be maintained, recovered, adapted or transformed, returning in this way to the initial system (in equilibrium) or to the system in different stages, i.e., with different levels of equilibrium, or even undergoing a change of state. Therefore, resilience assessment tools should be able to produce a set of improvements to the conditions of the system and identify challenges and opportunities that result from mitigation and prevention measures or the possibility of changing the functioning of the systems themselves, as proposed in Fig. 4.

7. Conclusions

For all its attractiveness and practical relevance, urban resilience is not a straightforward concept to be defined and put into practice. The research on the concept of urban resilience is increasing (Wang & Xue, 2019) according to different perspectives and scientific areas. This article reviews the scientific and technical literature on urban resilience, highlighting its definitions, dimensions, main characteristics, challenges and opportunities. Therefore, the paper aims to propose a conceptual framework to understand the urban resilience concept, in order to help governmental institutions to improve cities against disruptive events, which can affect their functioning.

Urban resilience consists on the capacity of a city and its urban systems to absorb the first damages, reduce the impacts arising from disturbances, adapt to change, and rapidly transform systems that limit current or future adaptive capacity. Nevertheless, the study of urban resilience, considering all kinds of disasters and disruptions in the systems of a city, is important to make communities more resilient when facing extreme events.

From the literature review, it can be concluded that urban resilience is mainly analyzed in terms of five dimensions, namely physical, natural, economic, institutional and social, which are supported by eleven basic characteristics: redundancy, diversity, efficiency, robustness, interdependence, adaptability, resources, independence, ingenuity, inclusion, integration, allowing the construction of an operational and implementable evaluation tool.

To promote more resilient urban systems, it is necessary to combine experiences from a variety of sources, including the integration of different types of knowledge and a variety of approaches in which scientists and public authorities can collaborate to generate reliable, contextualized, socially robust and implementable scientific knowledge. Thus, there is a need to integrate diverse areas of knowledge to produce an explicit operational definition and sound urban resilience baseline, overcoming the current levels of "imprecision" that are usually associated with urban resilience concept that varies according to the field of research. Therefore, a strategic and leadership structure is important to ensure the implementation of actions promoting the resilience of urban systems, since local management authorities are one of the main stakeholders.

Finally, despite the existence of many definitions, characteristics and dimensions used in the evaluation of urban resilience, it was possible to observe that the lack of tools to evaluate resilience creates the main challenge, and at the same time, the main opportunity for developing future studies in urban resilience. Therefore, it is necessary to identify the most effective types of measures to increase the resilience of different urban systems, how these should be designed, implemented and monitored. However, some recent literature (Leobons, Gouvêa Campos, & de Mello Bandeira, 2019; Martins, Rodrigues da Silva, & Pinto, 2019) showed that this topic is currently changing fast, at least in the case of transportation systems.

From the point of view of urban resilience, equilibrium in the functioning of a city cannot be easily replicated in other contexts, and this can also depend, among other aspects, on the resilience levels of the city's government system. This work can be the basis for future research or for the definition of improvement measures to the built urban environment of cities, allowing institutions (governmental or private companies) to promote actions or policies to mitigate and reduce the

impacts from natural hazards or human disruptive events. Thus, this work can be used as a political tool to plan new resilience policies applicable in cities.

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